

Chapter 17

Review for Midterm 2

Lecture 17 | Devore 9e: Ch. 3.5–4.5

Midterm 2 covers Lectures 12–16 — the remaining discrete families, continuous random variables, and the named continuous distributions — together with the hypergeometric distribution from Lecture 10, since Devore 3.5 treats it alongside the negative binomial. Note that joint distributions are *not* on this exam — they start next lecture.

17.1 What you must be able to do

1. **Change of variable.** Given X with known cdf/pdf and $Y = g(X)$, find the cdf and then the pdf of Y . Always go through the cdf; always state the new range.

$$F_Y(y) = \mathbb{P}\{Y \leq y\} = \mathbb{P}\{g(X) \leq y\} \longrightarrow f_Y(y) = F'_Y(y).$$

2. **Use a pdf.** Compute $\mathbb{P}\{a \leq X \leq b\} = \int_a^b f$, normalize an unknown constant via $\int f = 1$, and find $\mathbb{E}[X]$, $\mathbb{E}[h(X)]$, $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.
3. **Normal distribution.** Standardize, use the z -table for both probabilities and percentiles, exploit symmetry ($\Phi(-z) = 1 - \Phi(z)$).
4. **Derive** the negative binomial (Lecture 12) and hypergeometric (Lecture 10) pmfs — not just quote them. Both are counting arguments.
5. **Poisson process.** Know which of Poisson / exponential / gamma answers which question:

<i>how many</i> events in time t ?	Poisson(λt)
<i>how long</i> until the 1st event?	Exponential(λ) = Gamma($1, 1/\lambda$)
<i>how long</i> until the α -th event?	Gamma($\alpha, 1/\lambda$)

6. **Handle maxima and minima** (Lectures 8, 13, 15). When the variables come from independent trials, $\mathbb{P}\{\max \leq x\}$ is a product of cdfs and $\mathbb{P}\{\min > x\}$ a product of tails; for $X \geq 0$, $\mathbb{E}[X]$ is the integral of the tail. The first of several independent exponential waits is exponential, with the rates added.

17.2 Formula sheet

Continuous random variables

$$F(x) = \int_{-\infty}^x f(t) dt, \quad f(x) = F'(x), \quad \mathbb{E}[h(X)] = \int_{-\infty}^{\infty} h(x)f(x) dx,$$

$$\text{Var}(X) = \mathbb{E}[X^2] - \mu^2, \quad f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

Named continuous families

	pdf	$\mathbb{E}$, Var
Uniform $[A, B]$	$\frac{1}{B-A}$ on $[A, B]$	$\frac{A+B}{2}, \frac{(B-A)^2}{12}$
Normal (μ, σ^2)	$\frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$	μ, σ^2
Exponential (λ)	$\lambda e^{-\lambda x}, x \geq 0$	$\frac{1}{\lambda}, \frac{1}{\lambda^2}$
Gamma (α, β)	$\frac{x^{\alpha-1} e^{-x/\beta}}{\beta^\alpha \Gamma(\alpha)}, x \geq 0$	$\alpha\beta, \alpha\beta^2$
$\chi_k^2 = \text{Gamma}(\frac{k}{2}, 2)$	—	$k, 2k$
Weibull (α, β)	$\frac{\alpha}{\beta} \left(\frac{x}{\beta}\right)^{\alpha-1} e^{-(x/\beta)^\alpha}$	$\beta \Gamma(1 + \frac{1}{\alpha}), \beta^2 [\Gamma(1 + \frac{2}{\alpha}) - \Gamma(1 + \frac{1}{\alpha})^2]$
Lognormal (μ, σ^2)	$\frac{1}{\sqrt{2\pi}\sigma x} e^{-(\ln x - \mu)^2/2\sigma^2}$	$e^{\mu+\sigma^2/2}, (e^{\sigma^2} - 1)e^{2\mu+\sigma^2}$
Beta (α, β)	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$	$\frac{\alpha}{\alpha+\beta}, \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

Facts worth quoting

$$F_{\text{Exponential}}(x) = 1 - e^{-\lambda x}, \quad F_{\text{Weibull}}(x) = 1 - e^{-(x/\beta)^\alpha},$$

memorylessness: $\mathbb{P}\{X > s+t \mid X > s\} = \mathbb{P}\{X > t\}$ for the exponential only.

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}\{X > x\} dx \quad (X \geq 0), \quad \mathbb{E}[\max(Z, 0)] = \frac{1}{\sqrt{2\pi}}, \quad \mathbb{E}|Z| = \sqrt{\frac{2}{\pi}},$$

the minimum of independent $\text{Exponential}(\lambda_i)$ is $\text{Exponential}(\sum_i \lambda_i)$, and the k -th smallest of n independent $\text{Uniform}[0, 1]$ is $\text{Beta}(k, n-k+1)$, with mean $\frac{k}{n+1}$.

17.3 Five quick checks

Each should take about a minute. Answers follow.

1. Calls reach a help desk as a Poisson process with rate 30 per hour. What is the probability that a given minute passes with no call?
2. A component's lifetime is exponential with mean 2 years. It has already worked for 5 years. What is its expected remaining lifetime?
3. Which is larger, $\mathbb{P}\{Z > 1.645\}$ or $\mathbb{P}\{|Z| > 1.96\}$?
4. If $\ln X \sim \text{Normal}(0, 1)$, what are the median and the mean of X ?
5. Three numbers are drawn independently and uniformly from $[0, 1]$. What is the expected value of the smallest?

Answers.

1. The rate is $\frac{1}{2}$ per minute, so the probability is $e^{-1/2} \approx 0.61$.
2. 2 years. Memorylessness: the 5 years already survived are irrelevant.
3. Neither. Both equal 0.05: $1 - 0.95$ for the first, $2(1 - 0.975)$ for the second.
4. The median is $e^0 = 1$, since $\mathbb{P}\{X \leq 1\} = \mathbb{P}\{\ln X \leq 0\} = \frac{1}{2}$; the mean is $e^{1/2} \approx 1.65$.
5. $\frac{1}{4}$: the minimum of n uniforms has mean $\frac{1}{n+1}$ (Lecture 13).

17.4 The three mistakes that cost the most points

1. **Treating $f(x)$ as a probability.** It is a density; it may exceed 1, and $\mathbb{P}\{X = x\} = 0$ always.
2. **Losing the range in a change of variable.** A correct algebraic formula on the wrong interval is worth very little. Track where g maps the support of X .
3. **Mixing up λ and β .** The exponential is often written with rate λ ; the gamma with scale $\beta = 1/\lambda$. Write down which convention the problem uses before computing anything.

Remark 17.1. Practice problems: `exams/2026-spring/practice2.tex`, and the Fall and Spring Midterm 2 papers in `exams/`.